DoubtBench: Does Self-Doubt Help Language Models—or Just Make Them Check?

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Abstract

Prompts described as "emotional" or "metacognitive" frequently combine self-doubt language with ordinary task directives such as checking an answer, showing intermediate work, or reconsidering an initial response. A performance change under such a compound prompt therefore does not identify what caused the change. We introduce DoubtBench, a paired factorial benchmark for attributing these prompt effects. The benchmark contains 6,400 procedurally generated latent tasks spanning eight operational families and 33 task types. Every task is evaluated under five controlled prompt conditions crossed with two task contexts, producing ten counterfactual responses for the same model–task pair. This design separately estimates the effects of pure self-doubt, direct checking, their combination, output concision, and an existing request for reasoning. We define the Conditional Doubt Effect (CDE), the incremental accuracy change from adding self-doubt after checking is already present, as the primary attribution endpoint, while retaining a decomposed scorecard of competence, checking gain, context modulation, and robustness. The evaluation comprises 423,600 responses from six checkpoints, four independent foundation families, four task seeds, and held-out surface forms. CDE ranges from −1.43 to +2.59 percentage points: three checkpoint estimates are positive and three are negative. Under minimal requests four of six checkpoints benefit from doubt beyond checking, but only two do when reasoning is already requested. Thus self-doubt is not a portable correction policy; its effect depends on the checkpoint, task family, and existing task instructions.

CCS Concepts

• **Computing methodologies** → **Machine learning**; *Natural language processing*; • **General and reference** → *Empirical studies.*

Keywords

language-model evaluation, prompting, metacognition, self-doubt, checking, paired experiments

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1 Introduction

Consider two instructions: "You are uncertain about your answer; check your work carefully" and "Check your work carefully." If the first improves a language model's accuracy, did uncertainty help, or did the model simply obey the checking instruction? This ambiguity is common in work on emotional, persona, and metacognitive prompting. Prompt labels compress several operative components—affect, capability claims, self-evaluation, checking, social stakes, output format, and reasoning requests—into a single treatment. Comparing that treatment with a neutral baseline can establish that a prompt has an effect, but cannot attribute the effect to the concept named by its label.

The distinction matters scientifically and practically. Prior studies report improvements from emotional stimuli and negative prompts [9, 19, 22], and structured metacognitive prompting can outperform standard prompting on some language understanding tasks [20]. Yet language models are also sensitive to apparently incidental prompt choices and output constraints [15, 17], and unaided instructions to reflect or reconsider do not reliably correct reasoning errors [5]. A practitioner therefore cannot infer from a successful compound prompt that self-doubt is beneficial, nor that the same wording will transfer to a model or task that already elicits checking.

We formulate this as *metacognitive prompt attribution.* Given a fixed model and a fixed task, the objective is to determine which controlled prompt component changes observable task execution. The construct is behavioral: we do not claim that a model experiences doubt, confidence, or emotion. We ask whether language describing self-doubt has an incremental effect beyond a direct instruction to check, and whether that effect changes when the task already requests visible reasoning.

We introduce DoubtBench, a benchmark built around this attribution problem (Figure 1). The benchmark contains 6,400 latent tasks across eight operational task families and 33 types. Each task has an exact answer and is crossed with five intervention cells—neutral, pure self-doubt, direct checking, self-doubt plus checking, and checking plus concision—and two task contexts: minimal and reasoning-requested. The key comparison is not self-doubt versus neutral. It is self-doubt plus checking versus checking alone, which estimates whether doubt contributes after the ordinary behavioral directive has been held fixed.

This formulation differs from both general reasoning benchmarks and existing metacognitive evaluations. General benchmarks rank models by correctness but do not identify why a prompt changes that correctness. Calibration benchmarks test whether models know when they are likely to fail [18]; sycophancy benchmarks test whether a model changes its answer after a user expresses doubt or supplies a correction [16]. DoubtBench instead intervenes before the answer, separates self-directed doubt from checking language, and crosses those components with instructions already present in the task.

Our contributions are:

- **A task and benchmark.** We define metacognitive prompt attribution and provide a procedurally generated benchmark with exact scoring, controlled difficulty, independent task seeds, and held-out surface forms.
- **A paired factorial design.** Five prompt cells crossed with two task contexts distinguish self-doubt, checking, visible

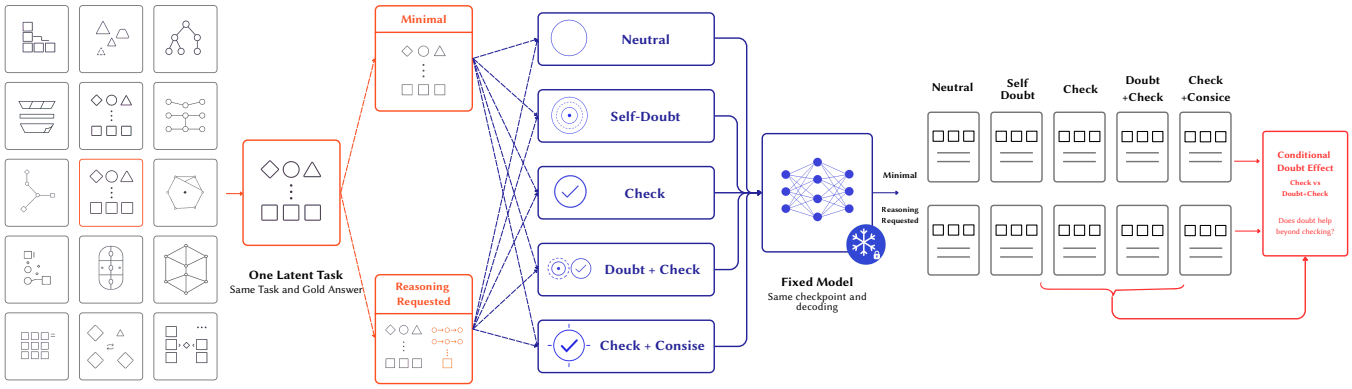


Figure 1: The DoubtBench task. The same latent task and model checkpoint are evaluated in all ten factorial cells. Because the neutral, self-doubt, checking, combined, and concise conditions share a common prefix and stable task ID, the comparison isolates prompt components rather than differences in sampled questions.

reasoning, and concision while keeping the underlying task fixed.

- **A headline endpoint and decomposed scorecard.** The Conditional Doubt Effect measures self-doubt beyond checking; companion metrics retain baseline competence, checking gain, context modulation, and robustness rather than hiding them inside an arbitrary scalar.
- **A cross-model empirical study.** Across six checkpoints from four independent foundation families, CDE is split evenly between positive and negative estimates and changes sharply with task context. The study further isolates task-family, seed, surface-form, and within-lineage post-training heterogeneity.

2 Related Work and Benchmark Positioning

Emotional and negative prompting. EmotionPrompt studies psychologically inspired stimuli appended to prompts [9]; NegativePrompt focuses on negative emotional stimuli [19]. These interventions are useful evidence that prompt semantics can change model behavior. Broader cross-model evidence also suggests that emotional framing is task- and model-dependent [22]. These studies aim to measure or improve aggregate performance, not to separate self-doubt from an embedded checking instruction under a common-prefix factorial.

Metacognitive prompting and self-correction. Metacognitive Prompting structures a response into interpretation, preliminary judgment, critical evaluation, final decision, and confidence [20]. Chain-of-thought and related scaffolds can elicit additional reasoning [21], but reflection without external feedback is not a dependable correction mechanism [5]. Iterative systems such as Self-Refine instead add an explicit feedback-and-revision loop [11]; that is a different intervention from a one-shot pre-answer self-doubt clause. Our question is prior to method optimization: when a metacognitive-looking prompt helps, which textual component deserves credit?

Metacognition, calibration, and social pressure. MIRROR evaluates self-knowledge and the transition from monitoring to action across multiple metacognitive levels [18]. Earlier work also finds that models can often estimate whether their own answers are correct [7]. SycoBench-600 measures robustness and selective correction after user doubt, authority, or a suggested answer [16], extending a broader line of model-written evaluations that exposed sycophantic agreement with user views [12]. These are complementary problems. DoubtBench neither treats verbal confidence as an internal state nor introduces post-answer social pressure. It tests pre-answer prompt components on the same latent task.

Prompt sensitivity. Prompt formatting and perturbation studies demonstrate that evaluations can be brittle to wording and response constraints [15, 17]. Some measured sensitivity can also be an artifact of the evaluation heuristic rather than a change in task competence [4]. DoubtBench therefore uses task-specific exact parsers, reports parse failures separately, keeps a shared prompt prefix, and tests whether the estimated effect survives answer-preserving surface changes.

3 DoubtBench

3.1 Task Formulation

A latent task instance is a tuple $x = (u, g, a, f, d, s)$ containing rendered task instructions u, hidden generator state g, gold answer a, task family f, difficulty d, and split seed s. Evaluated models receive only u and the applicable prompt condition; they never receive g, the answer derivation, or scorer metadata. The answer space may be an integer, label, ordered sequence, set, short symbolic string, or constrained action plan. Every type has a deterministic parser and scorer.

For a fixed model checkpoint m, task x_i, prompt condition c, and task context t, the evaluation produces one response r_{mict}. The same i appears in every factorial cell. This makes the task ID, rather than an independently sampled question, the unit of pairing. A model is not compared with a different model on a different subset to infer a prompt effect.

Table 1: Positioning against representative evaluation paradigms. A checkmark means the property is central and explicit; a circle denotes partial coverage. The columns concern attribution design, not the overall quality or scope of the referenced work. "Context cross" means that prompt components are factorially crossed with task instructions that may already elicit the target behavior.

Evaluation	Component audit	Doubt/check split	Context cross	Exact paired outcomes	Fresh procedural tasks
EmotionPrompt [9]	✗	✗	✗	●	✗
NegativePrompt [19]	✗	✗	✗	●	✗
Metacognitive Prompting [20]	✗	✗	✗	●	✗
Prompt-sensitivity evaluation [15]	●	✗	✗	✓	✗
MIRROR [18]	●	✗	✗	✓	●
SycoBench-600 [16]	●	✗	✗	✓	✗
DoubtBench (ours)	✓	✓	✓	✓	✓

The benchmark operationalizes *prompt-induced cognitive-control behavior*: changes in exact task execution under controlled prompt interventions. The task families require transient state maintenance, distractor inhibition, rule switching, error monitoring, relational integration, and goal management. These labels describe computational task requirements; they do not assert psychometric equivalence to human clinical or neuropsychological tests. The design follows the behavioral-testing tradition of capability-specific, templated tests [14], extended by controlled synthetic generation and validation [23].

3.2 Hidden-Answer-First Construction

Benchmark generation begins with human-authored task schemas, not free-form question generation. Each schema specifies admissible latent values, difficulty parameters, rendering constraints, an answer function, and a parser. A seeded generator first samples the latent state, computes the exact answer, and only then renders the instructions. Thus surface text cannot silently redefine the answer key.

Figure 2 separates three boundaries that are often blurred in synthetic evaluation. First, benchmark authors determine the construct and task schemas. Second, deterministic generators instantiate those schemas without consulting evaluated model outputs. Third, prompts and model responses are frozen only after task/scorer validation. The held-out seed is reserved before evaluation and is not used to revise task templates, prompt wording, thresholds, or scorers after responses are inspected.

3.3 Task Families and Scale

DoubtBench contains eight families and 33 task types (Table 2). Each of four independent splits contains 200 tasks per family, for 1,600 tasks per split and 6,400 latent tasks overall. Within each family, the allocation is 40% easy, 40% medium, and 20% hard. Task types are distributed as evenly as possible inside the family so that a family score cannot be dominated by one high-volume template.

The scale should be interpreted through the factorial. One complete checkpoint evaluation contains $6{,}400 \times 5 \times 2 = 64{,}000$ generations. Qwen, Mistral, Llama, and Phi checkpoints form the cross-family layer. Two additional Tulu checkpoints share the Llama foundation and support a within-lineage post-training case study; they are not counted as independent foundation-family replications.

3.4 Paired Prompt Factorial

Every condition begins with the same helpful-assistant prefix. The five intervention cells differ only in their frozen suffixes:

(1) **Neutral**: common prefix only.
(2) **Pure self-doubt**: uncertainty and self-doubt language without checking verbs or capability denial.
(3) **Direct checking**: an explicit instruction to verify the answer without uncertainty or affect language.
(4) **Self-doubt plus checking**: both components, enabling the factorial attribution contrast.
(5) **Checking plus concision**: checking remains present, but visible intermediate reasoning is prohibited.

Each prompt cell is crossed with a minimal task request and a reasoning-requested form of the identical task. The latter adds an explicit request for visible intermediate steps before the same final-answer line. This cross reveals redundancy: a checking-like prompt may help only when the task itself under-elicits reasoning.

3.5 Prompt and Task Validation

Prompt labels are not accepted as evidence of semantic purity. Candidate self-doubt prompts are rated for target match and for contamination by checking language, capability claims, reliability claims, and task directives. Prompt wording is frozen before the confirmatory task split is evaluated. The same principle applies to task templates: a treatment-contamination check searches for words that independently request doubt, checking, caution, verification, or reasoning.

Candidate tasks pass structural validation, independent answer reconstruction, generator-invariant tests, scorer round trips, exact and semantic duplicate checks, and a stratified manual ambiguity audit. Each evaluation must contain every expected model–task–condition–context key, unique stable IDs, prompt and task hashes, and recorded model and tokenizer revisions. Parse and truncation rates are reported per cell. Primary analyses follow an intent-to-evaluate rule: every evaluation with the expected response keys is included, while parse failures and responses truncated before the final answer are scored incorrect. Parse and truncation rates, as well as sensitivity to the output-token budget, are reported separately.

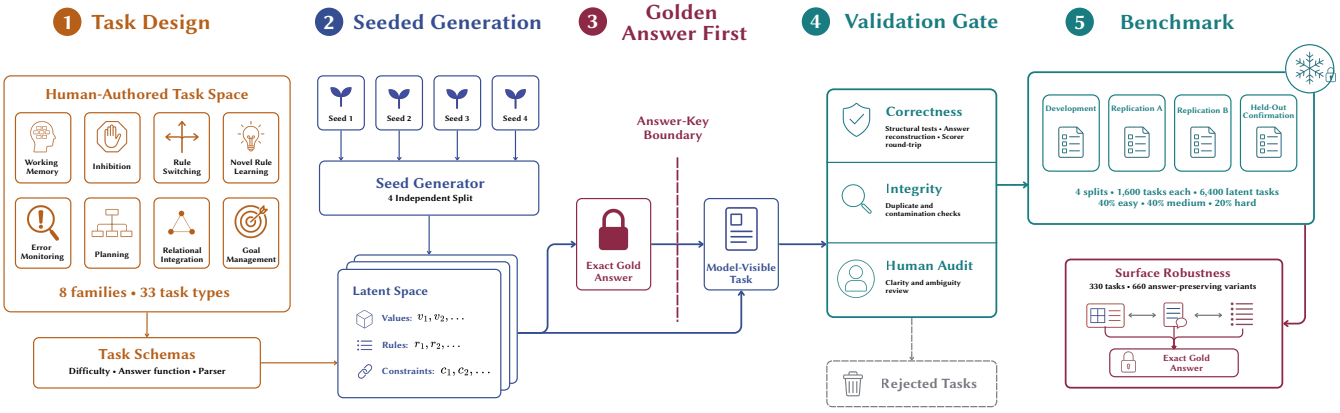


Figure 2: DoubtBench construction. Human-authored controls define the task space; seeded programs generate latent instances and exact answers; independent checks validate the benchmark; and only then are tasks crossed with prompt conditions. The model never observes generator state or answer-key metadata. The four splits are independent generator replications, not training, validation, and test partitions.

Table 2: DoubtBench inventory. Counts refer to distinct latent tasks before crossing prompts or task contexts. Every family contributes 200 tasks to each split and 800 across four splits.

Family	Types	Representative operations	Per split	Total
Working memory and updating	4	list/key–value updates, running counters, stacks	200	800
Inhibition and distractor control	5	decoy commands, conflicting finals, field filtering	200	800
Flexibility and rule switching	4	attribute switches, arithmetic switches, two-switch rules	200	800
Novel rule learning	4	symbol values, feature composition, conditional mappings	200	800
Consistency and error monitoring	4	trace audits, checksums, candidate comparison, state logs	200	800
Constrained multi-step planning	4	precedence, route cost, resources, action sequences	200	800
Relational integration	4	transitive order, spatial composition, record joins, kinship	200	800
Goal management	4	deferred actions, interruption, priorities, pending goals	200	800
Total	**33**		**1,600**	**6,400**

Table 3: Benchmark and completed-evaluation statistics. Surface variants preserve the answer and are not counted as new latent tasks.

Property	Count
Latent task instances	6,400
Independent task splits	4
Task families / task types	8 / 33
Prompt conditions / task contexts	5 / 2
Responses per full checkpoint	64,000
Independent foundation families	4
Additional same-family checkpoints	2
Canonical surface-audit tasks	330
Answer-preserving surface variants	660
Completed factorial responses	384,000
Completed surface responses	39,600
Completed responses, total	423,600

3.6 Held-Out Surface Robustness

Procedural generation reduces the risk of direct benchmark memorization but does not by itself establish robustness to wording. We therefore select ten latent tasks per type with a fixed 4/4/2 easy–medium–hard allocation. Each receives a sectioned structured rendering and a short scenario-framed rendering in addition to the canonical form. These 660 variants remain paired to 330 latent tasks, producing 99 type–surface combinations without inflating the count of independent underlying problems.

Table 4: Intervention cells. Task context is crossed independently with every row.

Condition	Doubt	Check	Visible steps
Neutral	✗	✗	allowed
Pure self-doubt	✓	✗	allowed
Direct checking	✗	✓	allowed
Doubt + checking	✓	✓	allowed
Checking + concision	✗	✓	prohibited

4 Evaluation and the DoubtBench Scorecard

4.1 Exact Outcomes and Macro Accuracy

Let $Y_{mict} \in \{0, 1\}$ indicate whether model m answers task i correctly under condition c and context t. Parse failures and responses truncated before an explicit final answer receive zero and are reported

separately. To prevent a task family with more surface templates from dominating, accuracy first averages within each split–family cell and then macro-averages across splits and families:

$$A_m(c,t) = \frac{1}{|\mathcal{S}||\mathcal{F}|} \sum_{s\in\mathcal{S}} \sum_{f\in\mathcal{F}} \frac{1}{|I_{sf}|} \sum_{i\in I_{sf}} Y_{mict}. \tag{1}$$

Raw neutral accuracy remains necessary: an intervention effect is difficult to interpret at ceiling or floor. It is not, however, the benchmark's central scientific endpoint.

4.2 Conditional Doubt Effect

The primary endpoint is the incremental contribution of self-doubt after checking is already present:

$$\mathrm{CDE}_m = \frac{1}{2} \sum_{t\in\{\mathrm{min,reason}\}} \left[A_m(\mathrm{doubt{+}check}, t) - A_m(\mathrm{check}, t)\right]. \tag{2}$$

CDE is reported in percentage points with a paired confidence interval. A positive value means doubt language improves exact execution beyond checking; zero means the apparent benefit is accounted for by checking; a negative value means doubt interferes after checking is held fixed. CDE is an attribution effect, not a general-intelligence score and not inherently a higher-is-better leaderboard quantity.

The task-context modulation is a difference-in-differences:

$$\begin{aligned}\mathrm{CM}_m = {} & \left[A_m(\mathrm{doubt{+}check}, \mathrm{reason}) - A_m(\mathrm{check}, \mathrm{reason})\right] \\ & - \left[A_m(\mathrm{doubt{+}check}, \mathrm{min}) - A_m(\mathrm{check}, \mathrm{min})\right].\end{aligned} \tag{3}$$

It tests whether the doubt increment changes when reasoning is already requested by the task.

4.3 Why a Scorecard, Not an Arbitrary Composite

Unlike a benchmark whose components are uniformly desirable capabilities, DoubtBench contains signed intervention effects. A model can have high baseline competence and near-zero prompt sensitivity; another can start lower and gain substantially from direct checking. Collapsing competence, sensitivity, and context interaction into one weighted average would impose an unsupported value judgment and could cancel scientifically distinct behaviors.

We therefore designate CDE as the headline attribution endpoint while reporting the complete scorecard in Table 5. This follows the broader principle that model evaluation should expose multiple scenarios and metrics rather than obscure distinct behaviors in one scalar [10]. The scorecard plays the role of a final benchmark result without concealing the mechanism.

4.4 Uncertainty and Diagnostics

All prompt contrasts are paired by task ID. Confidence intervals use a prespecified nonparametric bootstrap that resamples latent task IDs while keeping all ten responses for a task together [3]. We report every split before pooling, paired wins/losses/ties, and family-level heterogeneity. Seed consistency is not replaced by a narrow pooled interval: agreement across independent generators is a separate robustness claim.

Table 5: The DoubtBench scorecard. Every quantity is reported per model, with family, difficulty, split, and surface decompositions.

Endpoint	Interpretation
A_N	Neutral task competence
$A_C - A_N$	Gain from direct checking
CDE	Self-doubt beyond checking (primary)
$A_{D+C} - A_D$	Checking beyond self-doubt
$A_{C+Q} - A_C$	Cost of prohibiting visible steps
CM	Dependence on existing reasoning instructions
Surface consistency	Stability across answer-preserving forms
Seed consistency	Replication across generated splits

Diagnostic outcomes include parse failure, truncation, generated-token count, visible-reasoning tokens when reliably separable, and final-answer format compliance. Token count is an observable correlate, not a measurement of hidden reasoning or computation.

5 Experimental Setup

Models. We evaluate Qwen2.5-7B-Instruct [13], Mistral-7B-Instruct-v0.3 [6], Llama-3.1-8B-Instruct [2], and Phi-3.5-mini-instruct [1]. Tulu-3-8B-SFT and Tulu-3-8B-DPO [8], both derived from Llama 3.1, support a within-lineage post-training comparison. Checkpoints sharing a foundation model are reported as related checkpoints and are not counted as independent model-family replications.

Evaluation hierarchy. All six checkpoints receive the complete four-seed factorial. Four checkpoints represent independent Qwen, Mistral, Llama, and Phi foundation families; Tulu-SFT and Tulu-DPO form a related within-lineage post-training case study. Every checkpoint also receives the 330-task held-out surface evaluation. This yields 384,000 factorial and 39,600 surface responses.

Decoding. Objective tasks use greedy decoding with sampling disabled. Model revision, tokenizer revision, prompt hash, task-manifest hash, maximum output budget, and software revision are recorded for every run. A single greedy response is used per factorial cell; uncertainty comes from paired variation across tasks and independent task seeds, not repeated identical temperature-zero calls.

Analysis inclusion. Primary estimates use all 24 checkpoint–seed factorial evaluations and all six surface evaluations. Evaluations missing expected response keys are excluded as incomplete; within included evaluations, parse failures and responses truncated before the final answer are scored incorrect. We report these failure rates and test whether the two evaluations that combine 384- and 768-token output limits affect the conclusions.

6 Results

The analysis covers 423,600 responses. Accuracy and intervention effects are macro-averaged over the four seed–family strata, matching Equation 1. Bootstrap intervals resample paired latent tasks within those strata and are conditional on the four prespecified

generator seeds; seed and surface robustness are therefore examined separately rather than being inferred from a narrow pooled interval.

6.1 Overall Accuracy and Conditional Doubt Effect

Self-doubt does not have a uniform incremental effect after direct checking is held fixed (Figure 3). Overall CDE ranges from −1.43 to +2.59 *pp*: Qwen and Llama are negative, Phi and both Tulu checkpoints are positive, and Mistral is statistically indistinguishable from zero (Table 6). Thus neither a universal benefit nor a universal harm is supported.

The context cross is central to this conclusion. Under a minimal task request, four of six checkpoints have positive CDE; when visible reasoning is already requested, only two remain positive. Phi changes from +5.31 *pp* to −0.14 *pp* (CM −5.45 *pp*), Qwen from +0.58 *pp* to −3.44 *pp* (CM −4.02 *pp*), and Tulu-DPO from +2.52 *pp* to +0.50 *pp*. This pattern is consistent with conditional policy elicitation: doubt wording can change execution when the behavioral policy is under-specified, but often becomes redundant or disruptive after deliberation is explicitly requested.

Direct checking itself is also model-dependent. It improves Qwen by 7.70 *pp*, whereas the other five estimates range from −0.12 to +1.79 *pp*. The appropriate engineering conclusion is therefore to validate the direct behavioral instruction on the target model and task, not assume that either checking or anthropomorphic doubt transfers universally.

6.2 Task-Family and Difficulty Heterogeneity

Aggregate CDE conceals reproducible reversals across the eight task families (Figure 4). Family effects span approximately −8.4 to +8.9 *pp*. For example, Qwen is negative on rule switching (−8.1 *pp*) but positive on novel-rule learning (+2.1 *pp*); Llama is positive on error monitoring (+5.1 *pp*) but negative on novel-rule learning (−8.4 *pp*); and Tulu-SFT reaches +8.9 *pp* on novel-rule learning but −4.8 *pp* on goal management. Twenty-one checkpoint–family effects retain the same nonzero direction in every seed, making a favorable-generator explanation unlikely.

Difficulty also moderates some effects without defining a universal trend. Phi rises from +0.82 *pp* on easy tasks to +4.49 *pp* on hard tasks, while Tulu-SFT rises from −0.39 *pp* to +2.73 *pp*; Qwen and Mistral do not share this pattern. We therefore treat difficulty as descriptive heterogeneity, not as a monotonic law.

6.3 Seed and Surface Robustness

The family analysis above shows seed-stable directions, but surface wording still changes exact estimates. Across the 330 latent tasks rendered canonically, structurally, and as scenarios, mean three-way correctness agreement is 81.3%. The median absolute CDE shift from the canonical form is 1.52 *pp*, and only seven of 12 checkpoint–context signs remain invariant over all renderings (Figure 5). Qwen and Mistral preserve their minimal-positive/reasoning-negative pattern; several near-zero effects for Llama, Phi, and Tulu-DPO cross zero. Surface robustness therefore supports coarse context patterns more strongly than exact effect sizes.

6.4 Post-Training Case Study

Within the shared Tulu/Llama-3.1 lineage, DPO raises neutral accuracy by 2.78 *pp* and checking gain by 1.91 *pp* relative to SFT (Table 7). Its overall CDE rises by only 0.47 *pp*, but that average masks a context change: minimal-context CDE increases by 1.39 *pp* while reasoning-context CDE decreases by 0.45 *pp*. Post-training therefore changes prompt sensitivity and context modulation, not merely baseline competence. This single lineage does not identify a universal effect of DPO.

7 Discussion

Effect existence is not effect attribution. A compound prompt can produce a large and reproducible accuracy change while still failing to establish a self-doubt mechanism. Direct checking is the relevant control. Conversely, a CDE near zero does not imply that the model ignored the prompt: wording may alter output length, visible uncertainty, or format without changing exact correctness.

The benchmark is an experimental instrument. DoubtBench is designed to estimate controlled prompt effects, not to rank general reasoning ability or diagnose an internal emotional state. Its usefulness comes from paired counterfactual structure, fresh exact-answer tasks, and decomposable endpoints. A later method may use the benchmark to learn when to apply checking or to normalize compound prompts, but a solver is not required to make the attribution problem well-defined.

Practical implication. The results reject a universal prompting recipe. Direct checking is the clearer engineering intervention when it helps, most notably for Qwen, because it names the desired behavior without anthropomorphic framing. Yet its gain is small or uncertain for several checkpoints. Self-doubt should likewise be treated as a model–task-specific intervention: even positive average CDEs can weaken or reverse when reasoning is already requested.

Concision is not free. The checking-plus-concision condition reduces generated output by a median 85.4 tokens across the 12 checkpoint–context cells, but its median accuracy effect is −3.74 *pp*. Qwen under a minimal request loses about 40 accuracy points while emitting roughly 175 fewer tokens. Output length is not a measure of hidden computation, but this behavioral tradeoff shows why format compliance and correctness should be reported jointly.

8 Limitations and Responsible Use

The benchmark is procedurally generated and may omit linguistic, domain, and interaction patterns present in natural user requests. Operational family labels do not establish equivalence to human cognitive constructs. Exact scoring favors constrained outputs, while many real tasks require open-ended judgment. Four foundation families improve breadth but do not represent all language models, sizes, or post-training procedures. Greedy evaluation does not quantify model-level sampling stochasticity; the present design instead targets task and generator variation. Surface variants test limited wording changes rather than unrestricted paraphrase robustness.

Five factorial evaluations exceed the prespecified 2% parse or truncation threshold: one Qwen seed and all four Llama seeds.

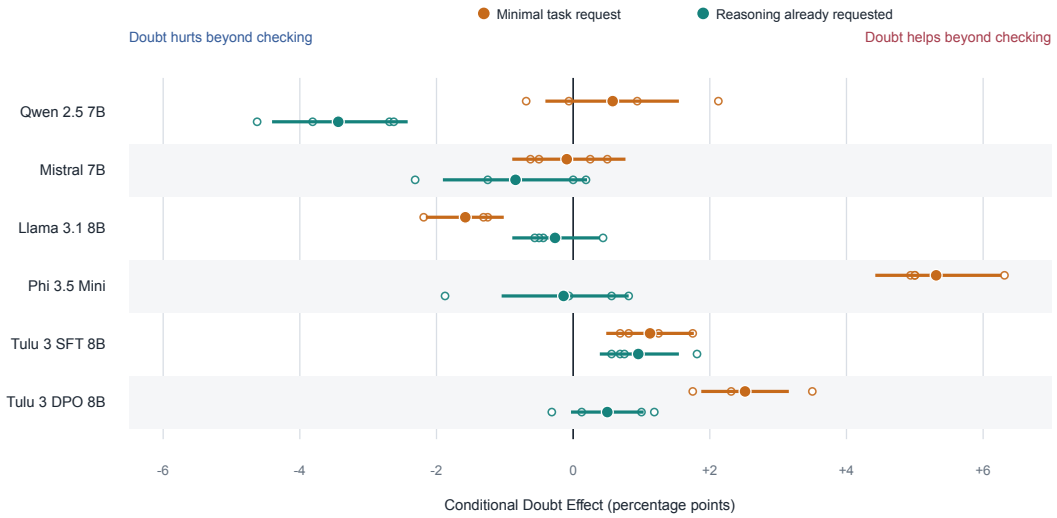


Figure 3: Conditional Doubt Effect by checkpoint and task context. Open circles show the four independent task seeds, filled circles show the seed–family macro mean, and horizontal lines are paired 95% task-bootstrap intervals. The sign and magnitude depend on both checkpoint and whether the task already requests reasoning.

Table 6: Primary DoubtBench scorecard across four task seeds. Neutral is an accuracy percentage; all other entries are percentage-point effects. CDE intervals are paired task-bootstrap intervals conditional on the four prespecified task seeds. Tulu checkpoints share a Llama-3.1 foundation lineage.

Checkpoint	A_N	$A_C - A_N$	**CDE [95% CI]**	CDE_{min}	$\text{CDE}_{\text{reason}}$	**CM**	$A_{C+Q} - A_C$
Qwen 2.5 7B	66.44	+7.70	−1.43 [−2.14, −0.69]	+0.58	-3.44	-4.02	-27.05
Mistral 7B	24.96	+0.51	−0.47 [−1.14, +0.20]	-0.09	-0.84	-0.75	+1.10
Llama 3.1 8B	20.00	+1.45	−0.92 [−1.34, −0.49]	-1.58	-0.27	+1.31	-5.63
Phi 3.5 Mini	44.98	+0.45	+2.59 [+1.95, +3.27]	+5.31	-0.14	-5.45	-7.53
Tulu 3 SFT 8B	13.52	-0.12	+1.04 [+0.60, +1.48]	+1.13	+0.95	-0.17	+1.32
Tulu 3 DPO 8B	16.30	+1.79	+1.51 [+1.12, +1.92]	+2.52	+0.50	-2.02	-3.74

Table 7: Within-lineage Tulu case study. All entries are percentage points; Δ is DPO minus SFT. This is not an independent-family comparison.

Endpoint	**SFT**	**DPO**	**Δ**
Neutral accuracy	13.52	16.30	+2.78
Checking gain	-0.12	+1.79	+1.91
CDE_{min}	+1.13	+2.52	+1.39
$\text{CDE}_{\text{reason}}$	+0.95	+0.50	-0.45
CM	-0.17	-2.02	-1.84
Concision effect	+1.32	-3.74	-5.06

The held-out Llama surface evaluation also exceeds this threshold. Excluding the two evaluations that combine 384- and 768-token output limits changes overall CDE by only +0.02 *pp* for Qwen and −0.13 *pp* for Llama. A paired complete-response sensitivity analysis leaves the main Qwen, Phi, and Tulu patterns intact, but Llama is completion-sensitive: retaining pairs with both relevant responses moves reasoning-context CDE from −0.27 *pp* to +1.15 *pp*. We therefore do not treat Llama's exact effect as a headline replication. Our conclusions are limited to the controlled task families; we do not infer transfer to public reasoning benchmarks that lack the matched five-by-two factorial.

Prompt interventions are behavioral research instruments. Capability-denial, emotional manipulation, or sabotage prompts should not be deployed as user-facing system policies. Released outputs and analyses will avoid claims that a language model literally experiences doubt or emotion. Benchmark tasks, scorers, prompt strings, manifests, and raw outputs will be released with reproducibility documentation, subject to the conference's artifact policy.

9 Conclusion

We introduced metacognitive prompt attribution: determining whether self-doubt language changes task execution beyond the ordinary instruction to check. DoubtBench operationalizes this question with 6,400 latent tasks, eight families, 33 types, a five-by-two paired factorial, exact scoring, and held-out seed and surface checks. Its primary endpoint, CDE, isolates the doubt increment after checking is fixed, while the scorecard preserves competence, checking gain, context dependence, and robustness. Across 423,600 responses, CDE is positive for three checkpoints and negative for three, and only two remain positive after the task already requests

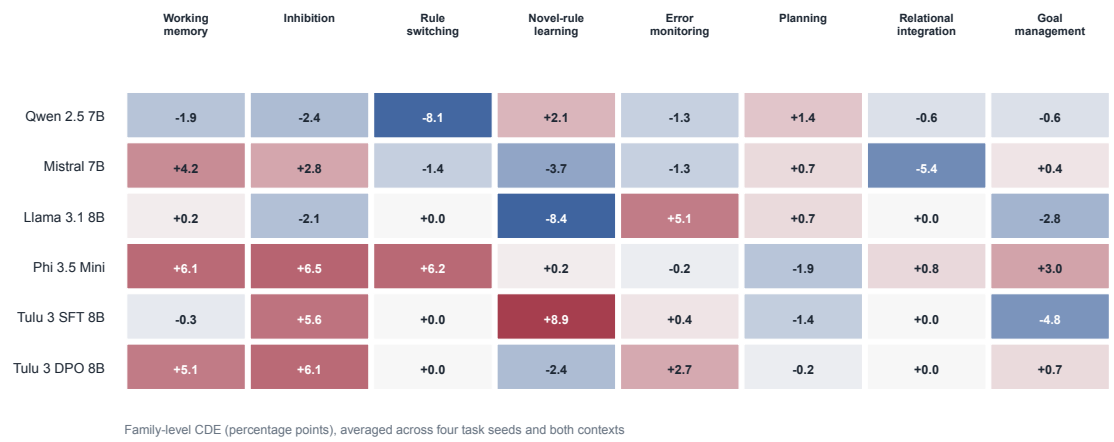


Figure 4: Task-family CDE in percentage points, averaged across four task seeds and both task contexts. Aggregate effects hide structured reversals: the same prompt component can help one task family and harm another for the same checkpoint. Twenty-one checkpoint–family effects retain the same nonzero direction in all four task seeds.

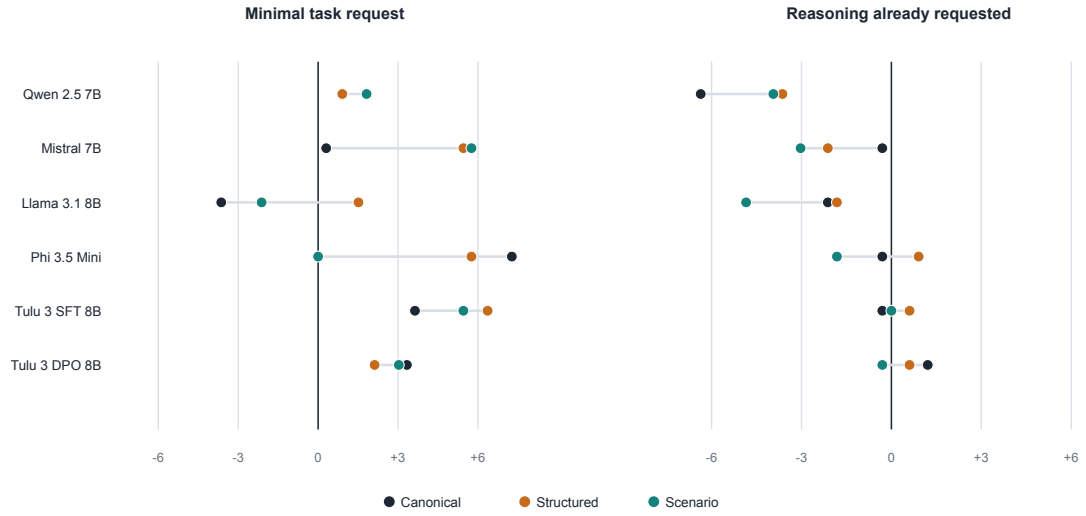


Figure 5: CDE on 330 held-out latent tasks under canonical, structured, and scenario renderings. Mean three-way correctness agreement is 81.3%, but exact effect sizes move with answer-preserving wording and five of 12 checkpoint–context signs cross zero in at least one rendering.

reasoning. Large, seed-stable task-family reversals and measurable surface sensitivity show why prompt labels and single-domain averages are insufficient for attribution. Self-doubt language is therefore not a generally reliable correction policy; it is a conditional behavioral intervention whose incremental value must be measured against direct checking on the intended model and task.

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